

# Fairness or Self-Interest? An Experiment on How Communication Shapes European Perceptions of Fairness in the EU Asylum Relocation System

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## Abstract

Despite intense political debate over reforming the EU asylum system, there is little evidence on what drives public support for alternative solutions. Using an experimental survey, we examine Europeans’ fairness perceptions of three mechanisms for distributing asylum seekers among EU countries: no relocation, relocation proportional to GDP, and relocation proportional to the resident populations. By varying the information provided about the consequences of these mechanisms, we explored the extent to which public perceptions are driven by concerns about outcomes versus principles of fairness. We find that most respondents (76%) consider relocation to be the fairest option. However, there are large cross-country differences, with proportional relocation schemes perceived as fair by 58% of respondents in Germany versus 87% in Greece, which suggests a “self-serving” bias — a tendency to rank as fairer the method that results in fewer asylum seekers for their own country. Moreover, when we communicate the expected number of asylum seekers for each relocation method, the self-serving bias intensifies significantly. By varying how the information is presented, we find a stronger effect when the communication displays *absolute* figures rather than figures *relative* to the population. These findings shed new light on how public perceptions of the fairness of the EU asylum system can be shaped by information.

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# 1 Introduction

Over the past years, the EU’s asylum and migration system has been severely tested, starting with the 2015 escalation of the Syrian war and continuing through more recent crises, such as Russia’s 2022 invasion of Ukraine. These events exposed a disproportionate pressure placed on Member States at the EU’s external borders. In response, the EU adopted the Pact on Migration and Asylum in May 2024, which establishes a permanent solidarity framework among Member States.

Under the Pact, the country of first entry remains responsible for asylum applications, while *all* Member States commit to support the system through solidarity contributions, including the relocation of applicants or beneficiaries, financial contributions, or other operational support.<sup>1</sup> However, there is no consensus among countries on how this responsibility should be shared. Because the implementation of solidarity mechanisms ultimately depends on political support at the national level, understanding public perceptions of fairness and solidarity in the EU asylum and migration system is crucial for ensuring the solidarity framework functions effectively and maintaining social cohesion across the EU.

To explore these perceptions, we conducted an experimental survey in eight EU countries. Respondents were asked to evaluate the fairness of alternative asylum-seeker relocation systems under different information conditions about the consequences of each mechanism. Specifically, the survey asked respondents to rank the fairness of three mechanisms: no relocation, allocation proportional to each country’s GDP, and allocation proportional to each country’s population.

As in Bansak et al. (2017), we compared responses from those who received information about the consequences of implementing each method for the respondent’s country against those with no such information. Extending this approach, we additionally vary how consequences are communicated, presenting them either in *absolute* terms or *relative* to the country’s population. This distinction is important because absolute figures may exaggerate perceived burdens, while relative measures better capture per-capita responsibility and may lead to different fairness judgments.

Our work builds on a large body of literature showing that people tend to combine several fairness principles, such as need, deservingness, or reciprocity, when judging distributional outcomes (Konow, 2003). However, when one’s own stakes are salient, a self-serving bias can shift those judgements so that the option that benefits agents the most appears “fairest” (Babcock et al., 1995; Konow, 2000). This tension between “distributional fairness” and “self-serving tendency” appears particularly relevant when considering international solidarity in the asylum system. On the one hand, the principle of fairness suggests that countries should share migrants proportionally to their capacity. On the other hand, self-serving biases may lead to prioritising national interest. However, despite pioneering work by Bansak et al. (2017) to understand this issue, how these views shape public perceptions of fairness in the migration system is still poorly understood.

Our analysis indicates that nearly three-quarters of participants perceive some form of within-EU relocation as a fair approach. However, there are significant country differences showing a tendency for participants to perceive as fairer relocation methods that result in fewer asylum seekers for their own country (“self-serving bias”). Alternative communication strategies can significantly moderate this tendency. Specifically, this study shows that presenting the likely consequences of different relocation rules in absolute terms — reporting how many asylum seekers are expected in the participants’ country

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<sup>1</sup> See European Commission (2024)

per year — significantly reinforces the self-serving bias. In contrast, presenting the same information in relative terms, rescaled by the residents’ population (i.e., asylum seekers per 100,000 inhabitants), appears to reduce this bias.

## 2 Related Literature

The EU Pact on Migration and Asylum is designed to manage the complex issue of migration more effectively. Its key component is the relocation system, aiming to create a fairer and more predictable process than the previous rules. The relocation of asylum seekers has been subject to increased controversy. This issue has fuelled significant political and social tensions for several reasons, including an uneven distribution whereby border countries bear most responsibility for processing asylum claims. Other EU member states were often unwilling to accept a share of these asylum seekers, leading to a major imbalance in burden-sharing (Thielemann, 2018). The issue has become a powerful tool for populist and far-right political parties across Europe, using anti-immigrant rhetoric to mobilise voters and gain political power (Dinas et al., 2019; Halla et al., 2017).

Despite an intense political debate about asylum-seekers’ relocation, there is still little evidence on Europeans’ preferences. Survey evidence indicates that Europeans favour capacity-based relocation over both the Dublin “first-entry” rule and financial transfers. The seminal paper of Bansak et al. (2017), using a survey with 18,000 respondents across 15 EU+ countries, shows that 72% of respondents backed proportional allocation even after learning it would raise their own country’s intake. Similarly, a 26-country experiment by C. W. Cappelen et al. (2023) showed that most people prefer admitting asylum seekers to paying host states. A cross-national conjoint study in eight EU countries found that citizens favour refugee protection paired with clear limits or conditions, rather than pure first-entry allocations (Jeannet et al., 2021). Existing research, hence, suggests that public preferences are more nuanced and conditional than commonly assumed. Recent UK-based findings by Corre and Tilley (2024) particularly highlight this complexity. They used a conjoint experiment to test whether current UK asylum policies align with public desire. Their results challenge the common belief that the public uniformly favours stricter policies than those currently in place. These results strongly suggest that to truly understand public preferences, researchers must disaggregate complex immigration policies into their specific components, moving beyond treating asylum as a single issue.

This debate can be framed within the discussion of the Equity Theory, originally formulated by Adams (1965). This framework posits that individuals evaluate the fairness of a situation by comparing the ratio of their own inputs and outcomes to those of others. In the context of asylum seekers redistribution, this translates into a demand for “proportionality,” where the perceived burden must be balanced against a country’s capacity or resources to maintain social and political equilibrium (Thielemann, 2003). When a distribution method is perceived as inequitable -e.g. when one country feels it is bearing a disproportionate share of the responsibility relative to its neighbours- it creates a state of distress, leading to policy resistance or non-compliance (Walster et al., 1978). Consequently, policy outcomes in the European Union often reflect an attempt to restore this perceived equity through the implementation of “distribution keys” based on objective criteria such as GDP and population size (European Commission, 2024). These mechanisms aim to align individual country preferences with collective outcomes.

Citizens’ widespread support for distributing asylum seekers based on a country’s capacity implies a

belief in the principle of proportional fairness and the equitable sharing of responsibility. However, this does not rule out that self-interest also shapes their attitudes, as some studies suggest. Bansak et al. (2017) found that the large majority of citizens favour an allocation policy proportional to each country's capacity over the status quo of allocation based on the country of first entry. This majority support, though weakened, persists even when respondents are informed that proportional allocation would increase the number of asylum seekers sent to their own country. These findings suggest that citizens care deeply about the fairness of the process.

Heizmann and Ziller (2020) analyses Eurobarometer data and finds that people living in countries that have taken in more asylum seekers and immigrants in the past are especially supportive of policies that redistribute wealth or resources, a preference they link to a mix of self-interest, concerns for their country's collective well-being, and moral values. Experimental work in Norway and Israel demonstrates that establishing clear proportional-allocation rules can mitigate NIMBYism (acronym for the "not in my back yard" attitude) by framing redistribution of refugees as a civic duty rather than an imposed burden (Solodoch, 2024). U.S. evidence from conjoint analysis further confirms that admission decisions are based more on collective good and ethical standards than on individual economic gain (Hainmueller and Hopkins, 2015).

Furthermore, a growing body of research confirms that public attitudes toward migration policy are malleable and can be meaningfully shifted through the use of concrete information and tailored framing. Studies show that specific informational framings can influence policy support: for example, informing U.S. respondents about the administrative burdens of legal immigration boosts support for more open policies (Kustov and Landgrave, 2025). Similarly, messaging congruent with personal values (using the Schwartz (1992) framework) proves more persuasive than generic humanitarian appeals in European campaigns (Dennison, 2020), and correcting misperceptions about immigration's labour-market impact increases and sustains public support (Haaland and Roth, 2020). In addition, narratives based on demographic challenges—such as focusing on labour shortages or an ageing population—can raise pro-immigration attitudes and political action (Facchini et al., 2022). Conversely, Bansak et al. (2017) found that providing consequences-focused information (specifically, revealing that a proportional policy would increase asylum seeker numbers in a respondent's own country) appeared to boost self-interest, thereby slightly weakening support for the fairness-based proportional relocation in those countries where proportionality and self-interest don't align. A. W. Cappelen et al. (2023) observed a self-serving bias where respondents deemed an allocation method fairer if it resulted in a lower national intake.

Drawing on the literature presented, we posit two core expectations (see the AEA RCT Registry for details). First, support for a proportional allocation system will be high: in each treatment, the share of participants considering the "no relocation" status quo as fairest will be significantly below the benchmark of one-third of all participants. Second, the perceived fairness of an option will negatively depend on the expected number of asylum seekers in one's own country under this option communicated to participants (self-serving bias). Specifically, consistent with research showing people overestimate migrant populations (Alesina et al., 2023), we expect self-serving bias to be more pronounced when the consequences are presented in absolute terms (thousands) compared to relative terms (hundreds per 100,000 inhabitants). This means that where a given relocation policy reduces the national burden with respect to the status quo (which is true for countries currently receiving a disproportionately high number of asylum applications), we predict that the policy will be rated as significantly fairer in the

information treatments. Conversely, in countries in which a proportional policy increases the national burden, we predict that the policy will be rated as significantly less fair in the information treatments. In all cases, we expect that the detrimental or beneficial self-serving effect will be strongest when the information is presented in absolute terms rather than relative terms. This prediction is based on the literature suggesting that the same social effect presented in absolute rather than relative terms might be perceived as larger, affecting policy preferences (Pedersen, 2017), a manifestation of the more general phenomenon of denominator neglect (Reyna and Brainerd, 2008).

This work contributes to the literature on how public perceptions of fairness influence policy acceptance. Policy acceptance across all sectors depends on citizens viewing them as fair. Fairness judgments about who deserves what, often guided by deservingness heuristics, are a central component of distributive justice and shape how citizens evaluate who should contribute and who should be supported under a given policy (Almås et al., 2010; C. W. Cappelen et al., 2023; Corneo and Grüner, 2002; Durante et al., 2014; Fong, 2001; Fong and Luttmer, 2011; Van Oorschot, 2006, 2000). Information can influence support, but its effects are heterogeneous: mobility information raises support (especially for opportunity policies), while inequality information mostly shifts concern and has large policy effects chiefly for the estate tax (Alesina et al., 2023; Kuziemko et al., 2015). In climate policy, support for carbon pricing rises when revenues are allocated in ways seen as fair, and acceptance can be improved with clear and trustworthy information on the policy’s impact (Carattini et al., 2019; Maestre-Andrés et al., 2019).

Our study also adds to the literature in behavioural economics and decision making, which aims to understand how people respond to different ways of framing information. Work in medical research, for instance, has shown that people respond differently to risks when they are presented in absolute versus relative terms (Malenka et al., 1993). Our work bridges these insights from medical and behavioural science literature and embeds them within a migration policy context.

## 3 Methods and Data

### 3.1 The survey

A sample representative of the adult population aged 18 years and over in eight EU countries (Bulgaria, France, Germany, Greece, Italy, Poland, Spain and Sweden) was drawn from Ipsos online panel by means of stratified random probability sampling. The selected countries were chosen to ensure a balanced representation of the European Union’s diversity, with particular attention to coverage across macro-regions (Northern, Southern, Eastern, and Western Europe). The selection also reflects variation in key structural characteristics, including economic conditions (GDP per capita), population size, and the share of foreign-born residents.

The sampling was stratified by age, gender and geographic region (NUTS 1). Pre-defined sub-sample sizes (“quotas”) were based on official population statistics provided by Eurostat. The overall sample size of the project included 8,122 participants. The main fieldwork took place between 30 September 2024 and 21 October 2024. Details of the field work and survey design are available elsewhere (see Blasco et al., 2025).

## 3.2 Experimental design

The survey embedded an experiment testing how different types of information affected participants' ranking of the fairness of three relocation rules for asylum seekers across EU countries:<sup>2</sup>

- **No relocation** (the baseline): asylum seekers remain in the country where they first arrive;
- Allocation **proportional to each country's population**: relocations ensure equity with population as a proxy for capacity;
- Allocation **proportional to each country's GDP**: relocations ensure equity with GDP as a proxy for capacity.

Respondents were asked to rank these allocation rules from the fairest (rank = 1) to the least fair (rank = 3). Before they did so, we varied the format and amount of information available across three conditions:

- **No information** (the control),
- **Absolute** information: the expected number of asylum applications under each relocation scheme (expressed in thousands),
- **Relative** information: the expected number of asylum applications under each relocation scheme per 100,000 citizens.

To illustrate the underlying relocation problem captured by our experiment, Table 1 reports the expected asylum applications in each country under each relocation rule. These figures are based on national statistics about each country's population, GDP, and historical (2014-2023) first-time asylum applications from Eurostat.<sup>3</sup>

A few observations can be made. First, historical asylum application rates vary substantially across countries, ranging from 16 per 100,000 inhabitants in Poland to 397 per 100,000 in Greece. Second, proportional allocation systems would significantly alter the current distribution of asylum applications. For example, under a population-proportional allocation scheme, Poland would receive an additional 153 applications per 100,000 inhabitants, while Greece would “send” 228 applications per 100,000 inhabitants to other countries. We refer to countries in this position as “net receivers” or “net senders.”

Third, proportional allocation rules would generate pronounced cross-country differences depending on the rule being based on population or GDP. For instance, under a GDP-proportional allocation system, Poland would still be a net receiver, but the additional inflow would be considerably smaller, decreasing from 153 applications per 100,000 inhabitants under a population-based system to 76 per 100,000 under a GDP-based system.

```
tbl_asylum_rel %>%  
  dplyr::mutate(  
    country = factor(  
      country
```

<sup>2</sup>The experimental design and pre-analysis plan were pre-registered within the AEA RCT Registry (AEARCTR-0014194). The Ethics Research Board of the JRC approved the experimental protocol.

<sup>3</sup>In this study, GDP and population are intended as proxies for a country's reception and processing capacity, consistent with existing EU allocation proposals and policy debates (Thielemann, 2018). These indicators also provide respondents with clear and easily interpretable signals of a country's size and wealth, which, as research shows, are commonly used when evaluating “fairness” in international sharing (Kleider and Stoeckel, 2019). Even so, respondents may interpret the proportionality by GDP or population in different ways (e.g., a proxy for housing capacity or labour market conditions), which may conflate the meaning of “capacity” and limit the interpretation of the results.

```

        country,
        levels = country[order(no_relocation, decreasing = TRUE)]
    ),
    d_pop = population - no_relocation,
    d_gdp = gdp - no_relocation
) %>%
dplyr::arrange(
    dplyr::desc(no_relocation)
) %>%
dplyr::transmute(
    Country = country,
    "No relocation" = no_relocation,
    "By population" = population,
    "By GDP" = gdp,
    `Δ Population` = d_pop,
    `Δ GDP` = d_gdp
) %>%
kableExtra::kbl(
    caption = paste(
        "Expected asylum applications under alternative",
        "burden-sharing rules (per 100,000 inhabitants)."
    ),
    booktabs = TRUE,
    linesep = "",
    align = c("l", rep("r", 5)),
    escape = FALSE
) %>%
kableExtra::kable_styling(
    latex_options = c("hold_position"),
    full_width = FALSE,
    position = "center",
    font_size = 10
) %>%
kableExtra::row_spec(0, bold = TRUE) %>%
kableExtra::add_header_above(
    c(" " = 1, "Levels" = 3, "Differences vs. no relocation" = 2),
    bold = TRUE
) %>%
kableExtra::footnote(
    general = "Entries report expected asylum applications per 100,000 inhabitants."
    threeparttable = TRUE,
    general_title = "Note:"
)

```



**Table 1:** Expected asylum applications under alternative burden-sharing rules (per 100,000 inhabitants).

| Country  | Levels        |               |        | Differences vs. no relocation |              |
|----------|---------------|---------------|--------|-------------------------------|--------------|
|          | No relocation | By population | By GDP | $\Delta$ Population           | $\Delta$ GDP |
| Greece   | 397           | 169           | 95     | -228                          | -302         |
| Sweden   | 344           | 169           | 230    | -175                          | -114         |
| Germany  | 313           | 169           | 219    | -144                          | -94          |
| Bulgaria | 178           | 169           | 65     | -9                            | -113         |
| France   | 151           | 169           | 184    | 18                            | 33           |
| Spain    | 138           | 169           | 136    | 31                            | -2           |
| Italy    | 128           | 169           | 158    | 41                            | 30           |
| Poland   | 16            | 169           | 92     | 153                           | 76           |

*Note:*

Entries report expected asylum applications per 100,000 inhabitants.

Finally, the survey included several modules that asked respondents about their socio-demographics, perceptions of the EU asylum system, support for different migration policies, and other background questions. The full questionnaire is in the Appendix. A descriptive analysis of the responses is presented in the technical report by Blasco et al. (2025). In our analysis, we included control variables such as trust or political affiliation, as these are associated with attitudes towards migration (e.g., Canetti et al., 2016; Mitchell, 2021). We also consider various socio-demographic factors, such as income and employment status, gender, age, and the parents' country of birth. Whether the respondent was born abroad may be relevant because foreigners may hold different opinions on migration issues than natives (Berlinschi and Harutyunyan, 2019).

### 3.3 Data

Table ?? in the Appendix illustrates the random assignment to the three information conditions in each country and overall. Each treatment group has about one third of total respondents per country, reflecting the stratified randomisation design.

Table 2 illustrates the distribution of key pre-treatment characteristics, including sex, age, education, trust, and perceptions across the three treatment groups. On average, respondents were 50 years old (with 16 years SD), with about half (47%) identifying as male. For simplicity of exposition and comparability across countries, education was merged into three broad categories: high (ISCED 5-8; 32% of respondents), medium (ISCED 3-4; 48%), and low (ISCED 0-2; 19%). On measures of trust, respondents reported moderate levels of general trust in others ( $M = 4.5$ ,  $SD = 2.7$ ), trust in the EU ( $M = 4.3$ ,  $SD = 2.7$ ), and trust in government ( $M = 3.7$ ,  $SD = 2.9$ ).

Knowledge of the EU asylum system averaged 2.0 ( $SD = 0.7$ ) on a 1-4 scale, thus the majority expresses no or basic knowledge. Most respondents reported there were “too many” asylum seekers in their country, with a mean of 2.6 on 1-3 scale from 1 = too few and 3 = too many. Similarly, on average, respondents view an unbalanced distribution of asylum seekers in the EU, reporting an average score of 3.1 ( $SD = 2.4$ ) on 0-10 scale for the question about evenness across the EU. Yet, respondents reported broad consensus on the support on the importance of burden sharing among countries, providing equal rights, equal treatment, and having an equal distribution of asylum seekers relatively high, with means ranging from 6.9 to 7.4 on 0-10 scales.



Table 2 also reports mean differences between the control group and the other treatment groups, with the standardised difference in parenthesis (the difference divided by the pooled standard deviation). For all these variables, the standardised differences between control and other groups is small, suggesting the distributions overlap well across treatment groups.<sup>4</sup>

```
# -----
# 1. Compute SMD
# -----
result <- lapply(names(ds), function(x) {
  column <- ds[[x]]

  if (!is.numeric(column) || all(is.na(column))) {
    return(NULL)
  }

  formula <- reformulate(c("relocation_treatment", "country"), x)
  fit <- lm(formula, data = ds, contrasts = list(country = "contr.sum"))

  ## compute pooled SD
  group_sds <- tapply(column, ds$relocation_treatment, sd, na.rm = TRUE)

  broom::tidy(fit) |>
    dplyr::mutate(
      pooled_sd = sqrt(sum(group_sds^2) / length(group_sds)),
      smd = estimate / pooled_sd,
      variable = x
    )
})

result <- dplyr::bind_rows(result)

# -----
# 2. Build table
# -----
tbl_covariates <- result %>%
  filter(
    str_detect(term, "relocation_treatment"),
    !str_detect(variable, "weight|restart_count|^t[DQ]|^tIn"),
  ) %>%
  mutate(
    stat = sprintf("%.2f [%].2f]", estimate, smd)
  ) %>%
  select(variable, term, stat) %>%
```

<sup>4</sup>These differences were estimated using OLS controlling for country fixed effects, thus accounting for the stratified randomisation.

```

pivot_wider(names_from = term, values_from = stat) %>%
mutate(
  variable = recode(variable,
    age = "Age (years)",
    male = "Male (%)",
    educ_high = "High education (%)",
    educ_mid = "Mid education (%)",
    educ_low = "Low education (%)",
    trust = "Trust others (0-10)",
    trust_eu = "Trust in EU (0-10)",
    trust_gov = "Trust in government (0-10)",
    political_right = "Political right (0-10)",
    share_legal_migrants = "Share legal migrants",
    asylum_knowledge = "Knowledge of asylum system (1-4)",
    asylum_fraud_detected = "Fraud detection ability (0-10)",
    asylum_decisions_objective = "Asylum decisions are objective (0-10)",
    asylum_distr_even = "Asylums in EU are even (0-10)",
    fair_share = "Asylums in own country (1 = too few, 3 = too many)",
    burden_sharing_important = "Importance of burden sharing (0-10)",
    equal_distribution_important = "Importance of equal distribution (0-10)",
    equal_treatment_important = "Importance of equal treatment (0-10)",
    equal_rights_important = "Importance of equal rights (0-10)",
    quick_processing_important = "Importance of quick process (0-10)",
    Duration = "Completion time (mins)",
    number_accepted_refugees = "Accepted refugees in own country",
    .default = variable
  ),
  group = case_when(
    variable %in% c(
      "Age (years)", "Male (%)",
      "High education (%)", "Mid education (%)", "Low education (%)") ~ "Sociodemographics",
    variable %in% c(
      "Trust others (0-10)", "Trust in EU (0-10)",
      "Trust in government (0-10)", "Political right (0-10)") ~ "Political attitudes",
    variable %in% c(
      "Knowledge of asylum system (1-4)",
      "Fraud detection ability (0-10)") ~ "Knowledge",
    variable %in% "Completion time (mins)" ~ "Survey administration",
    TRUE ~ "Other"
  )
)

```

```

) %>%
  arrange(desc(group)) %>%
  mutate(
    variable = str_replace(variable, "Importance of", "Value of"),
  )

groups <- unique(tbl_covariates$group)

tbl_covariates %>%
  mutate(variable = gsub("_", " ", variable)) %>%
  select(
    Variable = variable,
    `Relative treatment` = contains("Relative"),
    `Absolute treatment` = contains("Absolute")
  ) %>%
  kableExtra::kbl(
    booktabs = TRUE,
    align = c("l", "c", "c"),
    escape = TRUE,
    caption = paste(
      "Covariate balance across treatment arms.",
      "Mean differences with respect to the control group."
    )
  ) %>%
  kableExtra::kable_styling(
    full_width = FALSE,
    position = "center"
  ) %>%
  kableExtra::pack_rows(groups[1],
    start_row = 1,
    end_row = sum(tbl_covariates$group %in% groups[1])
  ) %>%
  kableExtra::pack_rows(groups[2],
    start_row = sum(tbl_covariates$group %in% groups[1]) + 1,
    end_row = sum(tbl_covariates$group %in% groups[1:2])
  ) %>%
  kableExtra::pack_rows(groups[3],
    start_row = sum(tbl_covariates$group %in% groups[1:2]) + 1,
    end_row = sum(tbl_covariates$group %in% groups[1:3])
  ) %>%
  kableExtra::pack_rows(groups[4],
    start_row = sum(tbl_covariates$group %in% groups[1:3]) + 1,
    end_row = nrow(tbl_covariates)
  )

```

**Table 2:** Covariate balance across treatment arms. Mean differences with respect to the control group.

| Variable                              | Relative treatment | Absolute treatment |
|---------------------------------------|--------------------|--------------------|
| <b>Survey administration</b>          |                    |                    |
| Completion time (mins)                | -1.11 [-0.06]      | -0.25 [-0.01]      |
| <b>Sociodemographics</b>              |                    |                    |
| Age (years)                           | -0.28 [-0.02]      | -0.15 [-0.01]      |
| Male (%)                              | 0.01 [0.03]        | 0.02 [0.03]        |
| High education (%)                    | -0.01 [-0.02]      | 0.01 [0.02]        |
| Mid education (%)                     | 0.00 [0.00]        | -0.01 [-0.01]      |
| Low education (%)                     | 0.01 [0.03]        | -0.00 [-0.00]      |
| <b>Political attitudes</b>            |                    |                    |
| Trust others (0-10)                   | 0.07 [0.03]        | 0.09 [0.03]        |
| Trust in government (0-10)            | 0.16 [0.06]        | 0.17 [0.06]        |
| Trust in EU (0-10)                    | 0.12 [0.04]        | 0.12 [0.04]        |
| Political right (0-10)                | -0.07 [-0.03]      | -0.07 [-0.03]      |
| <b>Other</b>                          |                    |                    |
| Asylums in EU are even (0-10)         | -0.10 [-0.04]      | -0.01 [-0.01]      |
| Value of burden sharing (0-10)        | -0.03 [-0.01]      | 0.03 [0.01]        |
| Value of equal distribution (0-10)    | -0.04 [-0.02]      | -0.02 [-0.01]      |
| Value of equal rights (0-10)          | -0.01 [-0.00]      | 0.07 [0.03]        |
| Value of equal treatment (0-10)       | -0.07 [-0.03]      | 0.04 [0.01]        |
| Value of quick process (0-10)         | -0.09 [-0.04]      | 0.02 [0.01]        |
| Accepted refugees in own country      | -65.91 [-0.00]     | 1159.07 [0.04]     |
| Asylum decisions are objective (0-10) | -0.06 [-0.02]      | -0.11 [-0.04]      |
| Share legal migrants                  | -0.20 [-0.01]      | 0.81 [0.03]        |
| Fraud detection ability (0-10)        | 0.09 [0.03]        | 0.12 [0.04]        |

*Note:*

Entries report OLS coefficients from regressions of each covariate on treatment indicators and country fixed effects. Standardized mean differences (SMDs) are reported in brackets.

```

) %>%
kableExtra::footnote(
  general = paste(
    "Entries report OLS coefficients from regressions",
    "of each covariate on treatment indicators and country fixed effects.",
    "Standardized mean differences (SMDs) are reported in brackets."
  ),
  threeparttable = TRUE,
  general_title = "Note:"
)

```

## 4 Discussion

This study used an experimental survey to examine Europeans' perceptions of the fairness of different mechanisms for distributing asylum applications among EU countries. By varying the information provided about the consequences of these mechanisms, it explored the extent to which public perceptions of fairness are driven by concerns about outcomes versus principles of fairness.

Consistent with prior research (Bansak et al., 2017), we find that a substantial majority (76%) of respondents across eight EU countries view some form of proportional relocation of applicants as *fairer* than no relocation at all. Moreover, most people view proportional relocation systems as *fairer* even when it would result in their own country receiving more asylum applications, as the share of those ranking relocation as fairer than no relocation exceeds 50% in all countries.

Yet, we find substantial variation in fairness perceptions across countries. In the control group, where no information about the outcomes was provided, we find that between 58% respondents in Germany to 87% respondents in Greece view some form of within-EU relocation as fairer than no relocation at all. This gap could be partly explained by concerns about the outcomes. For instance, a border country like Greece would experience more applications under a no relocation system, while Germany would experience fewer applications. This finding underscores the importance of disentangling concerns about outcomes from fairness.

Bansak et al. (2017) finds that varying the information available about outcomes significantly affects public support for different relocation systems. Building on this insight, our research demonstrates that providing information about the consequences is key in shaping public perceptions of the fairness of these systems. Thus, our study establishes a direct link between political support identified by Bansak et al. (2017) and the perceived fairness of different policy options. As such, our study adds to a growing literature exploring the role of public perceptions of fairness in regulation, such as Brick and Visser (2015) for climate regulation.

By providing information about the consequences of the allocation schemes, we find that when individuals learn about the outcomes, the initial preference for proportional schemes is offset if the relocation mechanism results in one standard deviation more asylum applications. In other words, Europeans are more likely to perceive proportional allocation systems as fair when the increase in the number of asylum seekers implied by these systems is not too large. A first key takeaway is that information matters: receiving information about the consequences of proportional mechanisms may alter perceptions of fairness, ultimately shaping public support for relocation policies.

Another key implication is that framing matters. Differences that appear large in absolute terms may seem small when presented in relative terms. Indeed, in our randomised experiment, we find that expressing the outcomes of different relocation systems in absolute numbers leads some individuals to react more strongly to the information. When no information is provided, a regime that increases expected asylum applications by one standard deviation is 13% less likely to be seen as fair. Once information is revealed, fairness drops by 40% with relative information and by 56% with absolute information. Thus, providing absolute information produces an additional 16-percentage point decrease in perceived fairness compared to relative information. This finding highlights another way in which public opinion can be influenced by the framing of identical information.

Our analysis of heterogeneous effects provides additional evidence on the mechanisms through which

absolute vs relative information works. These differing impacts are not uniform across the population but are highly dependent on individual factors, including political orientation and demographics. Across most education groups, we find a consistently negative and statistically significant correlation between the number of asylum seekers and perceived fairness. The magnitude of this association was larger in the absolute-information group, moderate in the relative-information group, and modest in the control group. However, the effect of asylum seekers was weakest for low-educated respondents in all groups. These findings suggest a minor sensitivity of low-educated individuals to information. With regard to political orientation, both GDP- and population-based relocation schemes were consistently viewed as fairer than no relocation, with the largest effects observed for left-leaning respondents under absolute and control conditions.

Our research relates to conspicuous experimental literature in other fields that has shown the sensitivity of human decision-making to absolute vs relative values (Kratochvíl and Havlíček, 2024; Malenka et al., 1993). In medical research, for example, effect sizes or risk reductions of interventions may appear very large or quite small, depending on the use of relative or absolute measures on the same dataset (Malenka et al., 1993). The use of relative risk reduction has been shown to be more likely to influence the decision to accept a treatment than absolute risk reduction. In political communication, the main reason why relative values are so widespread is that absolute values are mostly not of much use for comparison among countries because they vary in population size. It is worth noticing, however, that the use of relative information may lead to misleading conclusions (Silva, 2020). They should, therefore, be used when the measure of interest increases proportionally with population size. Our study provides additional evidence on the effect of relative vs absolute communication in the field of migration policies.

This study has some key limitations. As the evidence is survey-based, the findings rely on self-reported attitudes and are susceptible to biases such as social desirability or a gap between stated and actual behaviour. The scope of the research is also geographically constrained: drawing data from eight countries limits the generalizability of the results to global or even broader regional contexts. Finally, the use of hypothetical scenarios, though necessary for experimental control, means that the observed reactions to absolute versus relative information may not perfectly translate to real-world policy decisions.

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